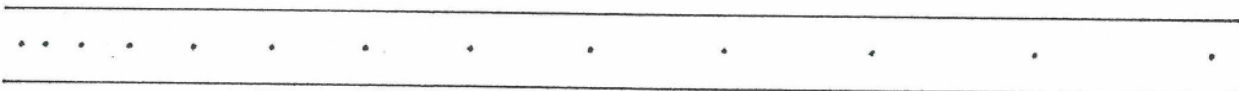


YEAR 10 SCIENCE (EXTENSION)
PHYSICS
REVISION

1. (a) Distinguish between the following terms.
 - (i) Distance and displacement.
 - (ii) Speed and velocity.
 - (b) A man drives his car 8.00×10^2 m due East for 2.10 mins and then turns North, driving 3.00×10^2 m in 1.35 mins before stopping at a petrol station.
 - (i) Calculate the average speed for the entire motion.
 - (ii) Calculate the displacement made from home to the petrol station. Show your working clearly.
 - (iii) Determine the average velocity for the entire motion.
2. A group of students produced the following ticker tape. One dot is recorded every 0.02 seconds.
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- (a) Describe the type of motion shown by the tape.
 - (b) Analyse the tape and calculate a value for the motion shown. Show your working.
3. A car moves off from a standing start with a uniform acceleration. It reaches 20.0 ms^{-1} after 8.00 s before maintaining its speed for another 15.0 s. As it approaches a set of lights, it brakes uniformly to 8.00 ms^{-1} over 12.0 s. It maintains this velocity as it moves through the lights, which turned green as it approached..
 (Assume the movement has occurred on a straight road.)
- (a) Draw a velocity - time graph for the entire motion.
 - (b) By finding the gradient, calculate:
 - (i) the acceleration at the start of the motion.
 - (ii) the deceleration as the car approached the second set of lights.
 - (c) From the graph, determine the displacement of the car over 40.0 s of the motion.
4. A balloon filled with hydrogen gas has a velocity of 40.0 ms^{-1} East. How long will it take to go 4.50×10^2 m ?

5. A hot air balloon is moving upwards at 5.00 ms^{-1} when $1.50 \times 10^2 \text{ m}$ above the ground. the pilot drops a 5.00 kg sand bag over the side to lighten the load and make the balloon rise faster.
- When the bag is released, does it move upwards or downwards initially? Explain your answer.
 - Calculate the velocity of the bag when it hits the ground.
 - How long does it take for the bag to reach the ground?
 - At its release, does the bag have potential energy (E_p), kinetic energy (E_k) or both E_p and E_k ? Explain your answer.
 - Calculate the total energy of the bag at its release.
 - Use your answer to part (e) above to calculate the velocity of the bag at impact with the ground.
 - Compare your answer with the answer to part (b). Explain your answer.
6. A car of mass $1.53 \times 10^3 \text{ kg}$ is travelling at 70.0 kmh^{-1} along Hepburn Avenue when the driver has to brake sharply to avoid a small child who suddenly appeared on a bike crossing the road. The car slowed to 20.0 kmh^{-1} in 3.20 s and managed to miss the child, who made it across safely.
- Calculate the average force exerted by the brakes.
 - How much work was done by the brakes in slowing the car?
(Hint: there are 2 ways to calculate an answer.)
 - The driver of the car was an engineer and had a hard-hat sitting on the back shelf. Describe what happened to the hard-hat in this situation and explain why it took place.
7.
 - Explain the difference between **mass** and **weight**. Give an example to help your explanation.
 - The acceleration due to gravity on Jupiter is 22.5 ms^{-2} . Calculate the weight of a $4.50 \times 10^2 \text{ kg}$ object on Jupiter.
 - A Canadian astronaut is working on the International Space Station. He is able to float freely about the cabin, moving easily about with little pushes off the walls. During a link-up with students from an Australian school, one student asked him what it felt like to be weightless. The student was surprised when he replied "I'm not weightless so I can't tell you". Was he correct to say this? Explain your answer.
8. A 70.0 kg cyclist is applying a force of $1.50 \times 10^2 \text{ N}$ forwards but is working against a wind producing a friction force of 80.0 N opposing the movement.
- Calculate the acceleration of the cyclist.
 - It is possible for the rider to move at constant velocity. In terms of the forces acting, explain how this is possible.
9. You are sitting on a chair at the moment. Explain why you don't fall through the chair and are supported by it.
10. Write the following in **scientific notation** and with **standard units (kg, m, s)**.
- | | | |
|------------------------------|-----------------------------|----------------------------|
| (a) 0.0293 mm | (b) 3652 kg | (c) 38.6 minutes |
| (d) 495 million km | (e) 70.0 kmh^{-1} | (f) 75 mg |

11. Perform the following calculation, expressing the answer to the correct number of significant figures.

$$\frac{(21.5+356.2)}{(1.19 \times 10^{-4})} = \frac{1}{2}(340)v^2$$

12. (a) What is a *scalar* and a *vector*?
(b) Give *two* examples of each.
13. A ball moves at 20.0 ms^{-1} due east for 1.00 s , collides with a wall and rebounds at 15.0 ms^{-1} south for 1.50 s . Calculate the average velocity of the ball.
(Hint: You need to consider vectors to solve this problem.)
14. Objects falling through the air reach *terminal velocity* (a point where they no longer accelerate). Explain why this happens.
15. A rocket accelerates upwards at 42.0 ms^{-2} from rest. It burns its motors for 3.25 s before coasting and deploying its parachute. However, the parachute is blown off the rocket as it is deployed, causing the rocket to plummet to the ground.
- (a) How high is the rocket when the motor finishes firing?
(b) What velocity would it have at this time?
(c) Which of Newton's Laws would best describe how a rocket works?
(d) What is the total time of flight for the rocket?
(Hint: You need to consider the time to return after the motor stops burning.)
16. (a) What force is required to accelerate an electron of mass $9.11 \times 10^{-31} \text{ kg}$ in an electric field from rest to $2.45 \times 10^7 \text{ ms}^{-1}$ over a distance of 3.00 cm ?
(b) How much energy does the electron have after the acceleration?
(c) How much work was done by the electric field?
(d) Consider the size of the force required to accelerate the electron. Are you surprised by its magnitude? Comment on this.
(e) Calculate the magnitude of the acceleration of the electron. Does this surprise you? Comment on this.
17. During a tennis match, a player lobs a ball from 1.35 m above the court with a vertical velocity of 11.0 ms^{-1} . It lands in the other court some 13.0 m distant. Ignoring any horizontal movement, calculate:
- (a) the height reached by the ball.
(b) the vertical velocity of the ball at impact with the ground.
(c) the time of flight.